Mark Rober's CrunchLabs My Experience Building the Mars Rover

Time	Subtitle	Human Translation
0s	If you look behind me, you'll see the center of the universe.	If you look behind me, you'll see the center of the universe.
3s	It's called the center of the universe	It's called the center of the universe
5s	because that's NASA's Jet Propulsion Laboratory.	because that's NASA's Jet Propulsion Laboratory.
7s	All of the orbiters, probes, and spacecraft	All of the orbiters, probes, and spacecraft
10s	we've sent out into the solar system and beyond	we've sent out into the solar system and beyond
13s	send all their signals right back here to be processed.	send all their signals right back here to be processed.
16s	That will never be more apparent than in a few days	That will never be more apparent than in a few days
18s	when the next Mars rover, Perseverance,	when the next Mars rover, Perseverance,
20s	concludes its seven-month journey to our neighboring red planet, Mars.	concludes its seven-month journey to our neighboring red planet, Mars.
24s	Autonomously navigating itself for a terrifying seven minutes,	Autonomously navigating itself for a terrifying seven minutes,
28s	traveling from 15 times the speed of a bullet	traveling from 15 times the speed of a bullet
31s	to a gentle three-miles-per-hour touchdown,	to a gentle three-miles-per-hour touchdown,
33s	while live-streaming the key data the whole way.	while live-streaming the key data the whole way.
36s	As many of you know, for me, this is like coming home.	As many of you know, for me, this is like coming home.
39s	Long before I started making YouTube videos,	Long before I started making YouTube videos,
41s	I came here to work every day for nine years,	I came here to work every day for nine years,
44s	seven of which were working on the last rover we sent to Mars, Curiosity.	seven of which were working on the last rover we sent to Mars, Curiosity.
48s	Today, we're gonna talk to some of my old friends	Today, we're gonna talk to some of my old friends
50s	and see the actual rover up close	and see the actual rover up close
52s	as I bring you up to speed on everything about this rover landing.	as I bring you up to speed on everything about this rover landing.
56s	Once you have an overview of what's gonna happen	Once you have an overview of what's gonna happen
58s	and what it took to get us to this point,	and what it took to get us to this point,
1:00	I feel certain you're gonna feel just as pumped	I feel certain you're gonna feel just as pumped

1:03	about this historic landing as I am.	about this historic landing as I am.
1:06	We're gonna talk about	We're gonna talk about
1:07	the who, what, why, where, and how of this rover.	the who, what, why, where, and how of this rover.
1:10	We'll start with the why and the where we're going.	We'll start with the why and the where we're going.
1:13	3.5 billion years ago, Earth and Mars were pretty similar.	3.5 billion years ago, Earth and Mars were pretty similar.
1:16	Both had liquid water on the surface	Both had liquid water on the surface
1:18	and both were protected from the sun's radiation with magnetic fields.	and both were protected from the sun's radiation with magnetic fields.
1:21	So it begs the question, if life first developed on Earth at that time,	So it begs the question, if life first developed on Earth at that time,
1:25	could it have also developed on Mars?	could it have also developed on Mars?
1:27	This is a massive lake in Jezero Crater billions of years ago.	This is a massive lake in Jezero Crater billions of years ago.
1:31	And this is it now.	And this is it now.
1:32	This is where Perseverance is landing.	This is where Perseverance is landing.
1:34	The bottom of an ancient lake the size of Lake Tahoe.	The bottom of an ancient lake the size of Lake Tahoe.
1:37	Using Earth as a guide, at the base of a river of fresh water	Using Earth as a guide, at the base of a river of fresh water
1:40	is where we have the best chance	is where we have the best chance
1:42	of finding evidence of past biological life on Mars.	of finding evidence of past biological life on Mars.
1:45	Thanks to Perseverance, we could be on the verge	Thanks to Perseverance, we could be on the verge
1:48	of the monumental first discovery of actual life outside our planet.	of the monumental first discovery of actual life outside our planet.
1:52	Being able to pinpoint a landing spot this tight	Being able to pinpoint a landing spot this tight
1:54	shows how NASA is constantly advancing its technologies.	shows how NASA is constantly advancing its technologies.
1:57	With Florida for scale,	With Florida for scale,
1:59	here's an oval showing the uncertainty of the landing spot	here's an oval showing the uncertainty of the landing spot
2:02	for previous missions Pathfinder in 1997,	for previous missions Pathfinder in 1997,
2:05	and then Phoenix in 2008, Curiosity in 2012,	and then Phoenix in 2008, Curiosity in 2012,
2:09	and now Perseverance.	and now Perseverance.
2:11	Being able to shrink a landing target gives you more options of places to land.	Being able to shrink a landing target gives you more options of places to land.
2:15	Plus, once you do land and start driving towards the actual science location,	Plus, once you do land and start driving towards the actual science location,

2:19	it could shave off a year or more of drive time.	it could shave off a year or more of drive time.
2:21	Not only does studying Mars help us understand Earth's past and future,	Not only does studying Mars help us understand Earth's past and future,
2:25	but the rovers are advance scouts,	but the rovers are advance scouts,
2:27	taking data and notes for us on the ground and sending all the info back to Earth.	taking data and notes for us on the ground and sending all the info back to Earth.
2:31	They're also testing new technologies that humans will need to use	They're also testing new technologies that humans will need to use
2:35	when we're exploring the planet ourselves in the very near future.	when we're exploring the planet ourselves in the very near future.
2:38	Because the first person to set foot on Mars is alive right now.	Because the first person to set foot on Mars is alive right now.
2:42	And it could be you.	And it could be you.
2:43	And if you're like, "But, Mark,	And if you're like, "But, Mark,
2:46	why would we spend resources and time exploring the solar system	why would we spend resources and time exploring the solar system
2:49	when we have big issues here on Earth we haven't solved?"	when we have big issues here on Earth we haven't solved?"
2:52	I tackle that question,	I tackle that question,
2:54	giving five reasons we can't afford not to invest in space	giving five reasons we can't afford not to invest in space
2:57	in another video you can watch after this one.	in another video you can watch after this one.
2:59	So that's the where and the why.	So that's the where and the why.
3:01	Let's talk about how we're going to do that.	Let's talk about how we're going to do that.
3:04	This is where it gets really wild.	This is where it gets really wild.
3:06	Meet Perseverance.	Meet Perseverance.
3:08	I should mention, I have my monthly videos planned out about a year in advance,	I should mention, I have my monthly videos planned out about a year in advance,
3:12	which is why exactly this time last year,	which is why exactly this time last year,
3:14	I knew I needed to fly down to check out the rover	I knew I needed to fly down to check out the rover
3:17	before it got shipped off to Florida to be launched.	before it got shipped off to Florida to be launched.
3:20	Before I checked out the rover,	Before I checked out the rover,
3:22	I stopped in to see Ben, who was my old boss.	I stopped in to see Ben, who was my old boss.
3:24	He was leading a small team	He was leading a small team
3:26	that designed the jetpack that lowered the rover to the ground.	that designed the jetpack that lowered the rover to the ground.
3:29	Now I heard he's all fancy, in charge of 400 people.	Now I heard he's all fancy, in charge of 400 people.

3:32	I wanted to get a sense of how things had changed for him.	I wanted to get a sense of how things had changed for him.
3:34	So anyone we see walk by, you can, like, boss them around?	So anyone we see walk by, you can, like, boss them around?
3:37	This guy? You can boss him around?	This guy? You can boss him around?
3:39	-Yeah, I can boss him around. -Okay, cool. Go on.	-Yeah, I can boss him around. -Okay, cool. Go on.
3:42	We first geeked out for a bit over parts from previous space missions.	We first geeked out for a bit over parts from previous space missions.
3:47	A hardware wall like this is a great illustration	A hardware wall like this is a great illustration
3:49	of what makes JPL such a cool place to work.	of what makes JPL such a cool place to work.
3:52	Actually, I designed this.	Actually, I designed this.
3:54	This is my hardware from GRAIL with Andy.	This is my hardware from GRAIL with Andy.
3:57	This was cool 'cause you got these flexures for temperature variation.	This was cool 'cause you got these flexures for temperature variation.
4:00	This isn't just for show.	This isn't just for show.
4:02	When you're figuring out a way to do things,	When you're figuring out a way to do things,
4:04	these are examples of how it's been done before.	these are examples of how it's been done before.
4:07	A lot are examples of the way you shouldn't do it.	A lot are examples of the way you shouldn't do it.
4:09	-That's why my-- -That's why your hardware's up there.	-That's why my-- -That's why your hardware's up there.
4:12	[Mark] We headed down to check out the rover and meet my friend Emily,	[Mark] We headed down to check out the rover and meet my friend Emily,
4:16	but before we could see it, we had to get suited up	but before we could see it, we had to get suited up
4:18	because the rover is looking for signs of biological life.	because the rover is looking for signs of biological life.
4:21	We don't wanna contaminate our samples before we even arrive.	We don't wanna contaminate our samples before we even arrive.
4:25	So a bunny suit and air shower can go a long way.	So a bunny suit and air shower can go a long way.
4:27	Emily was the vehicle-assembly lead for the descent and cruise stages,	Emily was the vehicle-assembly lead for the descent and cruise stages,
4:31	which is a big responsibility.	which is a big responsibility.
4:33	This is the rover.	This is the rover.
4:34	[Emily] The flight rover. It will be on Mars 12 months from now.	[Emily] The flight rover. It will be on Mars 12 months from now.
4:38	[Mark] It's so complex when you come up and get this close.	[Mark] It's so complex when you come up and get this close.
4:42	Perseverance is the most complex thing	Perseverance is the most complex thing

4:44	humans have ever built and sent to another planet.	humans have ever built and sent to another planet.
4:46	It's got laser, X-ray, and radar capabilities,	It's got laser, X-ray, and radar capabilities,
4:49	plus 19 cameras and a nuclear-powered battery system for energy.	plus 19 cameras and a nuclear-powered battery system for energy.
4:53	While it might look similar to the past rover, Curiosity,	While it might look similar to the past rover, Curiosity,
4:56	the science instruments are different because the objectives have changed.	the science instruments are different because the objectives have changed.
5:00	The most notable difference is	The most notable difference is
5:01	the drill isn't there just to create rock dust to study on the rover.	the drill isn't there just to create rock dust to study on the rover.
5:05	Perseverance has a hollow drill bit	Perseverance has a hollow drill bit
5:07	to core out a chunk of rock the size of a piece of chalk	to core out a chunk of rock the size of a piece of chalk
5:09	and then package it up and leave behind 43 separate samples	and then package it up and leave behind 43 separate samples
5:13	for a future mission to collect and send back to Earth.	for a future mission to collect and send back to Earth.
5:17	That way, we could study the samples for those past signs of life	That way, we could study the samples for those past signs of life
5:20	using state-of-the-art instruments that we could never fit on a rover.	using state-of-the-art instruments that we could never fit on a rover.
5:24	To capture that chalk-size rock sample,	To capture that chalk-size rock sample,
5:27	not only is there an arm on the outside, but there's one on the inside too.	not only is there an arm on the outside, but there's one on the inside too.
5:31	[woman] It is a miniature robot arm inside the body of the rover	[woman] It is a miniature robot arm inside the body of the rover
5:35	that manipulates the sample tube.	that manipulates the sample tube.
5:37	-There's one right here. -Wow.	-There's one right here. -Wow.
5:39	-So is this what you leave behind? -Exactly.	-So is this what you leave behind? -Exactly.
5:41	-This is, like, the poop of the rover? -Exactly. [chuckles]	-This is, like, the poop of the rover? -Exactly. [chuckles]
5:45	Okay. Do you like that analogy? They didn't like it earlier.	Okay. Do you like that analogy? They didn't like it earlier.
5:48	-Yeah. -You like it? Okay, cool.	-Yeah. -You like it? Okay, cool.
5:50	We like to say the rover's gonna poop out samples.	We like to say the rover's gonna poop out samples.
5:52	Okay, good. See? That's a good analogy.	Okay, good. See? That's a good analogy.
5:55	Perseverance is only the first leg of returning a piece of Mars to Earth.	Perseverance is only the first leg of returning a piece of Mars to Earth.
5:59	Future missions will complete what I call the "poop, scoop, and shoot" maneuver.	Future missions will complete what I call the "poop, scoop, and shoot" maneuver.
6:03	This is my friend Liz.	This is my friend Liz.

6:04	She's in charge of testing for the sample-retrieval system.	She's in charge of testing for the sample-retrieval system.
6:07	We're doing things that nobody else does, so we have to test it.	We're doing things that nobody else does, so we have to test it.
6:11	[Mark] She tests different configurations	[Mark] She tests different configurations
6:13	in a chamber that recreates the extreme temperatures and pressures	in a chamber that recreates the extreme temperatures and pressures
6:17	to make sure it will function on Mars.	to make sure it will function on Mars.
6:18	'Cause once you send something to space, it's gone, and you can't exactly fix it.	'Cause once you send something to space, it's gone, and you can't exactly fix it.
6:22	So it just has to work,	So it just has to work,
6:24	which is why testing is such a big deal here at JPL.	which is why testing is such a big deal here at JPL.
6:27	My buddy Matt mentioned another way this is done.	My buddy Matt mentioned another way this is done.
6:29	We build two spacecraft. One goes to Mars. One we keep to test.	We build two spacecraft. One goes to Mars. One we keep to test.
6:33	This is an exact replica of Curiosity, who's driving around on Mars right now.	This is an exact replica of Curiosity, who's driving around on Mars right now.
6:37	We use this one to test driving around obstacles and over rocks.	We use this one to test driving around obstacles and over rocks.
6:40	[Mark] And he knows about driving over rocks	[Mark] And he knows about driving over rocks
6:43	because he was a rover driver for both Opportunity and Curiosity.	because he was a rover driver for both Opportunity and Curiosity.
6:47	This guy is one of maybe 40 people in the world	This guy is one of maybe 40 people in the world
6:50	who has driven a vehicle on another planet,	who has driven a vehicle on another planet,
6:53	which is kind of a big deal.	which is kind of a big deal.
6:55	More cool things about Perseverance.	More cool things about Perseverance.
6:57	It has a mini helicopter drone	It has a mini helicopter drone
6:58	stowed away on its underbelly named Ingenuity.	stowed away on its underbelly named Ingenuity.
7:01	This will be mankind's first powered flight on another planet,	This will be mankind's first powered flight on another planet,
7:04	which makes this a Wright brothers moment.	which makes this a Wright brothers moment.
7:06	The rover and drone will get great footage, but we're mainly testing it out	The rover and drone will get great footage, but we're mainly testing it out
7:10	so that in the future, we might use drones to scout out terrain for us	so that in the future, we might use drones to scout out terrain for us
7:14	or get samples from hard to reach locations,	or get samples from hard to reach locations,

7:16	or you could have swarms of drones carrying materials for humans.	or you could have swarms of drones carrying materials for humans.
7:20	Perseverance is also testing a new instrument called MOXIE	Perseverance is also testing a new instrument called MOXIE
7:23	that basically amounts to a mechanical tree	that basically amounts to a mechanical tree
7:26	because its function is to convert CO2 into oxygen,	because its function is to convert CO2 into oxygen,
7:29	which future explorers will need to breathe and for rocket fuel.	which future explorers will need to breathe and for rocket fuel.
7:32	[Emily] The rover's been in this clean room for a year and a half,	[Emily] The rover's been in this clean room for a year and a half,
7:35	starting as just a chassis, just the skeleton.	starting as just a chassis, just the skeleton.
7:38	[Mark] All the teams and engineers have been taking turns	[Mark] All the teams and engineers have been taking turns
7:41	building up their part until it's done.	building up their part until it's done.
7:43	I was in that position on Curiosity,	I was in that position on Curiosity,
7:46	designing my hardware for three and a half years,	designing my hardware for three and a half years,
7:48	and when it was all tested and complete,	and when it was all tested and complete,
7:50	integrating it on the rover here in this room.	integrating it on the rover here in this room.
7:53	For the other three and a half years,	For the other three and a half years,
7:55	I was working on a small team of engineers on the jetpack descent stage.	I was working on a small team of engineers on the jetpack descent stage.
7:59	We've covered where and why we're going.	We've covered where and why we're going.
8:02	Also, the how we're gonna do that with the rover.	Also, the how we're gonna do that with the rover.
8:04	Now let's talk about the what.	Now let's talk about the what.
8:06	What's gonna happen when it lands and what you should expect to see.	What's gonna happen when it lands and what you should expect to see.
8:09	As I edit this video, the spacecraft is gliding toward Mars	As I edit this video, the spacecraft is gliding toward Mars
8:12	at a cool 48,144 miles per hour.	at a cool 48,144 miles per hour.
8:16	How fast is that? It's this fast.	How fast is that? It's this fast.
8:18	It's 15 times faster than a bullet.	It's 15 times faster than a bullet.
8:20	It's traveling 100 soccer fields in exactly this long.	It's traveling 100 soccer fields in exactly this long.
8:24	It will keep on that trajectory until the big moment	It will keep on that trajectory until the big moment
8:27	when it starts its entry, descent, and landing, or EDL.	when it starts its entry, descent, and landing, or EDL.
8:30	It's also known as the Seven Minutes of Terror	It's also known as the Seven Minutes of Terror

8:32	because we've got seven minutes	because we've got seven minutes
8:35	to get from the top of the atmosphere to the surface of Mars,	to get from the top of the atmosphere to the surface of Mars,
8:38	going from 13,000 miles per hour to zero in perfect sequence and perfect timing,	going from 13,000 miles per hour to zero in perfect sequence and perfect timing,
8:43	and the spacecraft has to do it on its own with no help from us on Earth.	and the spacecraft has to do it on its own with no help from us on Earth.
8:46	When it hits the upper atmosphere,	When it hits the upper atmosphere,
8:48	friction causes the heat shield to start glowing like the surface of the sun.	friction causes the heat shield to start glowing like the surface of the sun.
8:52	All the while, thrusters are firing	All the while, thrusters are firing
8:54	to steer and adjust its course towards the target location.	to steer and adjust its course towards the target location.
8:57	And that aerobraking gets rid of 99% of the energy,	And that aerobraking gets rid of 99% of the energy,
9:00	so for the last 1%, we deploy a supersonic parachute.	so for the last 1%, we deploy a supersonic parachute.
9:03	Then we pop off the heat shield we no longer need	Then we pop off the heat shield we no longer need
9:06	so the radar can start viewing the ground.	so the radar can start viewing the ground.
9:08	But even with the parachute, it's traveling 200 miles per hour,	But even with the parachute, it's traveling 200 miles per hour,
9:12	which is way too fast to land.	which is way too fast to land.
9:13	So we cut loose of the backshell and fire the rockets.	So we cut loose of the backshell and fire the rockets.
9:17	We can't land in this configuration	We can't land in this configuration
9:18	because the rockets will kick up debris and damage the rover.	because the rockets will kick up debris and damage the rover.
9:22	So we lower it from a 21-foot rope and gently land the rover on the surface	So we lower it from a 21-foot rope and gently land the rover on the surface
9:26	as my sky crane zooms off to face an honorable, catastrophic ending	as my sky crane zooms off to face an honorable, catastrophic ending
9:30	as far away from the rover as its fuel will carry it.	as far away from the rover as its fuel will carry it.
9:32	So in just seven minutes, the spacecraft has completely metamorphosized,	So in just seven minutes, the spacecraft has completely metamorphosized,
9:36	shedding all its sacrificial elements	shedding all its sacrificial elements
9:39	until you're left with just a rover, sitting alone,	until you're left with just a rover, sitting alone,
9:42	safely on the surface of Mars.	safely on the surface of Mars.
9:43	Everything you just saw was a CGI animation,	Everything you just saw was a CGI animation,
9:46	but a few days after landing,	but a few days after landing,
9:47	we'll all be blown away by actual HD landing footage	we'll all be blown away by actual HD landing footage

9:50	from the 23 cameras and two microphones on board.	from the 23 cameras and two microphones on board.
9:53	We'll see the parachute inflate and hear the crunch of the wheels	We'll see the parachute inflate and hear the crunch of the wheels
9:56	as they touch down and make contact with the Martian surface.	as they touch down and make contact with the Martian surface.
10:00	And because Mars is so far away,	And because Mars is so far away,
10:02	to get a signal from the vehicle to our planet,	to get a signal from the vehicle to our planet,
10:04	it takes about 12 minutes at the speed of light.	it takes about 12 minutes at the speed of light.
10:07	That means after the spacecraft sends a signal	That means after the spacecraft sends a signal
10:09	that it's reached the atmosphere,	that it's reached the atmosphere,
10:11	by the time that signal reaches Earth to kick off the Seven Minutes of Terror,	by the time that signal reaches Earth to kick off the Seven Minutes of Terror,
10:15	for at least five minutes,	for at least five minutes,
10:17	the vehicle has actually already been on the surface,	the vehicle has actually already been on the surface,
10:20	either alive or dead.	either alive or dead.
10:22	Which is why it has to be autonomous.	Which is why it has to be autonomous.
10:23	That means it makes its own choices on the timing of things and where to steer	That means it makes its own choices on the timing of things and where to steer
10:27	without anyone controlling it,	without anyone controlling it,
10:29	which means all we can do is watch and monitor and hope.	which means all we can do is watch and monitor and hope.
10:33	And as easy as these dedicated teams of engineers make it look,	And as easy as these dedicated teams of engineers make it look,
10:37	landing on Mars is really hard.	landing on Mars is really hard.
10:39	Historically, only about half the attempts have been successful,	Historically, only about half the attempts have been successful,
10:42	but the willingness to take big risks to reap big rewards	but the willingness to take big risks to reap big rewards
10:45	is the foundation on which NASA is built.	is the foundation on which NASA is built.
10:47	The livestream to watch the landing starts on Thursday, February 18th	The livestream to watch the landing starts on Thursday, February 18th
10:50	at 11:15 a.m. Pacific.	at 11:15 a.m. Pacific.
10:52	It will hit the top of the atmosphere an hour and a half later at 12:48 p.m.,	It will hit the top of the atmosphere an hour and a half later at 12:48 p.m.,
10:56	which starts the Seven Minutes of Terror, and we touch down at 12:55.	which starts the Seven Minutes of Terror, and we touch down at 12:55.
11:00	I'll leave a link in the video description as well as some other really cool sites,	I'll leave a link in the video description as well as some other really cool sites,
11:05	such as this video-game-like demo where you can interactively experience	such as this video-game-like demo where you can interactively experience

11:08 the Seven Minutes of Terror in preparation for landing. the Seven Minutes of Terror in preparation for landing.

11:11 We covered where and why we're going, we covered the how with the rover, We covered where and why we're going, we covered the how with the rover,

11:15 and the what with the landing details. and the what with the landing details.

11:17 Now it's time for the who. Now it's time for the who.

11:18 The human side to sending robot explorers to other planets. The human side to sending robot explorers to other planets.

11:23 What makes NASA and JPL great aren't its robots. It's the people who build them. What makes NASA and JPL great aren't its robots. It's the people who build them.

11:28 While it's not possible While it's not possible

11:29 for me to capture what it's gonna feel like for them to see it land, for me to capture what it's gonna feel like for them to see it land,

11:32 I can show you what it felt like for me eight years ago I can show you what it felt like for me eight years ago

11:35 when I was in a similar situation. when I was in a similar situation.

11:37 I made a video when Curiosity landed, I made a video when Curiosity landed,

11:39 but at the time, my channel had less than 100,000 subscribers, but at the time, my channel had less than 100,000 subscribers,

11:42 so I know most of you haven't seen this. so I know most of you haven't seen this.

11:44 Even if you have, with landing less than a week away, Even if you have, with landing less than a week away,

11:47 it's a great time for a second watch to get you pumped up. it's a great time for a second watch to get you pumped up.

11:50 For context, you should know that during my time working on the rover, For context, you should know that during my time working on the rover,

11:54 I had a son and lost my mom to ALS. I had a son and lost my mom to ALS.

11:56 While my contribution was small relative to the overall picture, While my contribution was small relative to the overall picture,

12:00 I tried to capture the human element here I tried to capture the human element here

12:02 of what it feels like to have seven years of your life and career of what it feels like to have seven years of your life and career

12:06 vindicated with that beautiful phrase, vindicated with that beautiful phrase,

12:08 "Touchdown confirmed. We are safe on Mars." "Touchdown confirmed. We are safe on Mars."

12:11 So consider this my tribute to everyone out there So consider this my tribute to everyone out there

12:14 working to push the limits of human understanding, working to push the limits of human understanding,

12:16 with a little bit of help from adorable little young Mark Rober. with a little bit of help from adorable little young Mark Rober.

12:24 We're still two days from landing, We're still two days from landing,

12:25 but you can see the news media has already started to descend on JPL. but you can see the news media has already started to descend on JPL.

12:32	Since we're all nerds here, our office pools look different.	Since we're all nerds here, our office pools look different.
12:36	This is, uh, the landing ellipse for the rover,	This is, uh, the landing ellipse for the rover,
12:39	and we all place guesses on where we think it's gonna come down.	and we all place guesses on where we think it's gonna come down.
13:12	[man] Heading for the target.	[man] Heading for the target.
13:18	Cruise stage separation.	Cruise stage separation.
13:46	[man] We have seen heating of the heat shield through the MEDLI instrument.	[man] We have seen heating of the heat shield through the MEDLI instrument.
13:55	Dynamics, Phase. Come back again with, uh…	Dynamics, Phase. Come back again with, uh…
14:01	[man] Parachute deployed.	[man] Parachute deployed.
14:11	We are in powered flight. Standing by for sky crane.	We are in powered flight. Standing by for sky crane.
14:17	Sky crane has started.	Sky crane has started.
14:34	-Touchdown confirmed. We're safe on Mars. -[cheering]	-Touchdown confirmed. We're safe on Mars. -[cheering]
14:37	That's touchdown!	That's touchdown!
14:39	Whoa!	Whoa!
14:44	Woo-hoo!	Woo-hoo!